

ASSIGNMENT : 14

ADVANCED PROGRAMMING

NAME-BHITALI BORUAH

ROLLNO-CSB24045

SESSION-SPRING

YEAR-2024-28

```
EXPLORER
...
assignment13.c
assignment13.exe
assignment14.py

assignment14.py > ...
1  import gc
2  import sys
3
4  class Node:
5      def __init__(self, name):
6          self.name = name
7          self.link = None
8
9      def __del__(self):
10         print(f"{self.name} is deleted")
11
12  gc.enable()
13
14  A = Node("A")
15  B = Node("B")
16
17  A.link = B
18  B.link = A
19
20  print("Reference count of A:", sys.getrefcount(A))
21  print("Reference count of B:", sys.getrefcount(B))
22
23  a_id = id(A)
24  b_id = id(B)
25
26  del A
27  del B
28
29  print("\nDeleted A and B")
30
31  found = False
32  for obj in gc.get_objects():
33      if id(obj) == a_id or id(obj) == b_id:
34          found = True
35          print("Objects still exist in memory due to circular reference")
36
37  collected = gc.collect()
```

```
EXPLORER
...
assignment13.c
assignment13.exe
assignment14.py

assignment14.py > ...
38
39  print("\nGarbage Collector ran")
40  print("Unreachable objects collected:", collected)
41
42  found_after = False
43  for obj in gc.get_objects():
44      if id(obj) == a_id or id(obj) == b_id:
45          found_after = True
46
47  if not found_after:
48      print("Cycle cleaned successfully")
```

OUTPUT-

```
assignment14.py > ...
38
39     print("\nGarbage Collector ran")
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    SPELL CHECKER 2
c:\Users\VITU\OneDrive\Desktop\c>python -u "c:\Users\VITU\OneDrive\Desktop\c\assignment14.py"
Reference count of A: 3
Reference count of B: 3

Deleted A and B
Objects still exist in memory due to circular reference
Objects still exist in memory due to circular reference
A is deleted
B is deleted

Garbage Collector ran
Unreachable objects collected: 2
Cycle cleaned successfully

c:\Users\VITU\OneDrive\Desktop\c>
```